

News Article Classification (Fake/Real)

Introduction

In the era of digital information, the spread of fake news has become a serious challenge. The aim of this project is to build a machine learning-based classifier that can differentiate between real and fake news articles. This project utilizes Natural Language Processing (NLP) techniques to process news content and develop a reliable classification system.

Abstract

This project focuses on classifying news articles as fake or real using NLP. We collected a labeled dataset containing both fake and genuine news from Kaggle. The text data was preprocessed using NLTK tools to remove noise and standardize the content. After vectorization with TF-IDF, machine learning models such as Logistic Regression trained. The classifier achieved high accuracy and F1-score, demonstrating its ability to detect fake news effectively. A Streamlit-based UI was also developed to allow users to input news content and receive instant predictions with explanations.

Tools Used

- Programming Language: Python
- Libraries: Pandas, NLTK, Scikit-learn
- Modeling: Logistic Regression
- Interface: Streamlit
- Evaluation: Accuracy

Steps Involved in Building the Project

1. Data Collection
 - A labeled dataset was obtained from Kaggle containing both fake and real news articles.
2. Data Cleaning & Preprocessing
 - Used NLTK to remove stopwords, punctuation, and apply lemmatization.
 - Converted all text to lowercase and removed irrelevant tokens.
3. Feature Extraction
 - Applied TF-IDF vectorization to convert cleaned text into numerical feature vectors for model training.

4. Model Training

- Trained models using Logistic Regression and.
- Used a train-test split to validate performance.

5. Model Evaluation

- Evaluated models using accuracy, precision, recall, and F1-score.
- Logistic Regression gave better overall performance with high F1-score.

6. User Interface

- Developed an interactive Streamlit app for real-time news classification.
- User can paste news content to get prediction.

Conclusion

The project successfully implements a fake news detection system using machine learning and NLP. It highlights the importance of text preprocessing, feature extraction, and model selection in building an effective classifier. With a simple and functional UI built on Streamlit, the system allows real-time fake news detection and can be extended further for multi-class classification or integration with web APIs.